



PEOPLES INFORMATION TECHNOLOGY PROGRAM
QUEST CAMPUS, NAWABSHAH



DATABASE MANAGEMENT SYSTEMS
MIDTERM EXAMINATION

DATE: DEC 19, 2024,

Time Allowed: 2 hours

Maximum Marks: 50

PART A: Multiple choice questions

30 Marks

Section I: MySQL (Basics, CRUD, SELECT, WHERE, Joins, Triggers, Procedures)

1. What is the primary key in a MySQL table used for?

- A) To ensure all values in a column are unique
- B) To allow NULL values
- C) To define a default value for a column
- D) To create relationships between tables

2. Which SQL statement is used to retrieve data from a table?

- A) INSERT
- B) UPDATE
- C) SELECT
- D) DELETE

3. What does the WHERE clause do in an SQL query?

- A) Filters rows based on a condition
- B) Joins two tables
- C) Groups rows based on a column
- D) Sorts rows in ascending order

4. Which SQL keyword is used to sort the result set in ascending order?

- A) ORDER BY
- B) SORT BY
- C) GROUP BY
- D) ASC

5. How would you select all records where the salary is between 30,000 and 50,000?

- A) SELECT * FROM employees WHERE salary = 30000 OR 50000;
- B) SELECT * FROM employees WHERE salary BETWEEN 30000 AND 50000;
- C) SELECT * FROM employees WHERE salary >= 30000 && salary <= 50000;
- D) Both B and C

6. Which of the following is an aggregate function in SQL?

- A) COUNT
- B) CONCAT
- C) LEFT
- D) NOW

7. What does the JOIN operation do in MySQL?

- A) Deletes duplicate records
- B) Combines rows from two or more tables based on a related column
- C) Creates a new database
- D) Updates the schema of a table

8. Which JOIN type returns all rows from both tables, even if there is no match?

- A) INNER JOIN
- B) LEFT JOIN
- C) RIGHT JOIN
- D) FULL OUTER JOIN

9. If `SELECT * FROM orders`; retrieves 100 rows, what will `SELECT COUNT(*) FROM orders`; do?

- A) Retrieve all rows
- B) Return the number 100
- C) Return the primary key values
- D) Delete duplicate rows

10. What is a trigger in MySQL?

- A) A scheduled event in MySQL
- B) A function that executes when certain conditions are met in the database
- C) A query optimizer
- D) A column constraint

11. When does a `BEFORE INSERT` trigger execute?

- A) After data is inserted into the table
- B) Before data is inserted into the table
- C) When a table is created
- D) Before data is updated in the table

12. Which SQL command is used to call a stored procedure?

- A) EXECUTE
- B) CALL
- C) RUN
- D) TRIGGER

13. How can you import a database into MySQL Workbench?

- A) By writing all queries manually
- B) Using the "Data Import/Restore" option
- C) By exporting data first
- D) By using a SELECT query

14. What happens if you try to insert a duplicate value into a column with a `UNIQUE` constraint?

- A) The row is inserted successfully
- B) The row is rejected with an error
- C) The database crashes
- D) The duplicate value is stored with a warning

15. What does the following query return?

```
SELECT department_id, COUNT(*) FROM employees GROUP BY department_id HAVING COUNT(*) > 5;
```

- A) Departments with more than 5 employees
- B) Total count of all employees
- C) Departments with exactly 5 employees
- D) Count of employees in all departments

Section II: MongoDB Basics

16. In MongoDB, what is a collection?

- A) A row in a database
- B) A document within a database
- C) A group of related documents
- D) A set of commands

17. Which data format is used by MongoDB to store documents?

- A) JSON
- B) XML

C) SQL

D) CSV

18. Which command is used to insert a new document into a MongoDB collection?

A) ``db.collection.add()`` B) ``db.collection.insertOne()``

C) ``db.collection.update()`` D) ``db.collection.push()``

19. What does the ``find()`` method do in MongoDB?

A) Deletes a document

B) Updates a document

C) Retrieves documents from a collection

D) Creates a new collection

20. How would you update the field ``age`` to 30 for all users whose ``name`` is "John"?

A) `db.users.find({name: "John"}, {$set: {age: 30}})`

B) `db.users.updateMany({name: "John"}, {$set: {age: 30}})`

C) `db.users.update({name: "John"}, {age: 30})`

D) `db.users.replaceOne({name: "John"}, {age: 30})`

21. Which of the following commands deletes a specific document in MongoDB?

A) `db.collection.removeOne()`

B) `db.collection.deleteOne()`

C) `db.collection.dropOne()`

D) `db.collection.removeDocument()`

22. What is the purpose of the `_id` field in MongoDB?

A) To store timestamps

B) To uniquely identify each document

C) To index all fields in the document

D) To define the schema of the document

23. What is the default storage engine used in MongoDB?

A) RocksDB

B) WiredTiger

C) InnoDB

D) MyISAM

24. How can you display all databases in MongoDB?

A) `show tables`

B) `show collections`

C) `show databases`

D) `db.list()`

25. Which MongoDB operator is used to match a range of values?

A) `$range`

B) `$lt` and `$gt`

C) `$in`

D) `$exists`

Section III: MONGODB AND MYSQL

26. Which of the following is a NoSQL database?

A) MySQL

B) MongoDB

C) Oracle

D) PostgreSQL

27. What is the primary difference between MySQL and MongoDB?

A) MySQL stores data in tables, MongoDB in documents

B) MongoDB requires structured schemas, MySQL does not

C) MySQL is faster than MongoDB in all cases

D) MongoDB does not support CRUD operations

28. Which of the following commands is used to export a MongoDB database?

- A) mongoexport
- C) mongoimport

- B) mongodump
- D) mongobackup

29. What is a FOREIGN KEY in MySQL?

- A) A key that uniquely identifies rows in its table
- B) A key used to reference a column in another table
- C) A key that enforces uniqueness in a column
- D) A key that increases query performance

30. What is the purpose of the \$set operator in MongoDB?

- A) To delete a field from a document
- B) To update or add fields in a document
- C) To insert a new document
- D) To filter documents

PART B: Subjective Questions**20 Marks**

Q. 01 : What is a database? Explain its purpose and how it is different from a traditional file-based system.

Q. 02: Why do we need normalization in relational databases? Explain the main objectives and benefits of normalization.